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Carcinogenesis of Male Oral Submucous Fibrosis Alters Salivary Microbiomes

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Abstract

Most oral squamous cell carcinoma (OSCC) tumors arise from oral premalignant lesions. Oral submucous fibrosis (OSF), usually occurring in male chewers of betel quid, is a premalignant stromal disease characterized by a high malignant transformation rate and high prevalence. Although a relationship between the inhabited microbiome and carcinogenesis has been proposed, no detailed information regarding the oral microbiome of patients with OSF exists; the changes of the salivary microbiome during cancer formation remain unclear. This study compared the salivary microbiomes of male patients with OSCC and a predisposing OSF background (OSCC-OSF group) and those with OSF only (OSF group). The results of high-throughput sequencing of the bacterial 16S rRNA gene indicated that OSF-related carcinogenesis and smoking status significantly contributed to phylogenetic composition variations in the salivary microbiome, leading to considerable reductions in species richness and phylogenetic diversity. The microbiome profile of OSF-related malignancy was associated with increased microbial stochastic fluctuation, which dominated the salivary microbiome assembly and caused species co-occurrence network collapse. Artificial intelligence selection algorithms consistently identified 5 key species in the OSCC-OSF group: *Porphyromonas catoniae, Prevotella multisaccharivorax, Prevotella* sp. HMT-300, *Mitsuokella* sp. HMT-131, and *Treponema* sp. HMT-927. Robust accuracy in predicting oral carcinogenesis was obtained with our exploratory and validation data sets. In functional analysis, the microbiome of the OSCC-OSF group had greater potential for *S*-adenosyl-L-methionine and norspermidine synthesis but lower potential for L-ornithine and pyrimidine deoxyribonucleotide synthesis and formaldehyde metabolism. These findings indicated that the salivary microbiome plays important roles in modulating microbial metabolites during oral carcinogenesis. In conclusion, our results provided new insights into salivary microbiome alterations during the malignant transformation of OSF.

Keywords: precancerous conditions, microbiota, dysbiosis, machine learning, saliva, biomarker

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Introduction

Oral squamous cell carcinoma (OSCC) is a burden globally, particularly in Asia (Petersen et al. 2005). In Taiwan, OSCC is also a socioeconomic concern because patients with OSCC are significantly younger than those with other malignancies (Huang et al. 2015). Meanwhile, oral submucous fibrosis (OSF) is a well-known oral stromal premalignant disorder, usually occurring in male chewers of betel quid (Lee et al. 2003; Yang et al. 2017). It can affect most parts of the oral cavity and sometimes the oropharynx (Rajendran 1994). In OSF, the textures of the mucosa, epithelium, and stroma become fibrotic, hypocellular, and hypovascular (Tekade et al. 2017), respectively. According to global evidence (Lian et al. 2013), OSF is accompanied by an atypically high malignant transformation rate (odds ratio, 23.3 to 27.5) as compared with that in healthy people (Yang et al. 2017), as well as rapid oral cancer formation (Lian et al. 2013). Moreover, most patients with OSF have mouth-opening limitation, making it difficult to thoroughly evaluate the oral cavity (Cheng et al. 2014). Hence, early detection and treatment for these high-risk patients with OSCC-OSF remain crucial clinical challenges.

Invasive biopsy is the gold standard for diagnosing OSCC. Currently, no effective noninvasive tool for screening large populations exists, particularly for patients with OSF. Considerable efforts have been exerted to identify biomarkers for OSF carcinogenesis, but most biomarkers require invasive specimen collection, followed by immunochemistry or tissue microarrays (Sharma and Radhakrishnan 2017). Prior research verified saliva as a source of biomarkers for detecting OSCC, but no effective saliva biomarker has been identified for patients with OSF. Because saliva is readily accessible for noninvasive collection and analysis, identifying biomarkers in it to predict and detect OSCC in patients with OSF can facilitate early intervention in this high-risk population.

A growing body of evidence has revealed the close association of microbiota dysbiosis with human diseases (Proctor et al. 2019). The oral cavity features complex habitats for the growth of diverse bacterial populations, such as the mucous surfaces of the buccal regions, tongue, gingiva, and tooth surfaces. The oral bacterial community is complex, consisting of >700 bacterial species dynamically colonizing specialized oral microenvironments (Aas et al. 2005). The diversity, composition, and functionality of this microbiome are altered by many diseases, including oral cancer (Lee et al. 2017; Zhao et al. 2017). However, no data are available regarding the saliva microbiome signatures of OSCC formation in the OSF background. Habitual betel quid chewing has been reported to alter oral microbiomes in cohorts of healthy individuals and patients with OSCC (Hernandez et al. 2017; Zhong et al. 2020). To minimize the influence of potential confounding effects, this study focused on addressing the relationship between the oral microbiome and OSF carcinogenesis in male chewers of betel quid. We comprehensively compared the salivary microbiomes of male patients with OSF alone (OSF group) and male patients with OSCC and a predisposing OSF background (OSCC-OSF group) to explore the role of microbiomes in OSF progression to OSCC.

Materials and Methods

All participants were recruited from the Department of Stomatology, National Cheng Kung University Hospital, and the Chi-Mei Medical Center, both in Tainan, Taiwan, with approval from the Institutional Review Boards of both institutions (Appendix Table 1). Saliva samples were obtained from the OSF ($n = 18$) and OSCC-OSF ($n = 34$) groups. All participants were men who used betel quid, and they had not received antibiotics or surgical treatments for at least 1 mo prior to enrollment. Participants were interviewed and asked to refrain from eating, drinking, or using oral hygiene products for at least 1 h before saliva collection. Following a previously reported protocol for saliva collection (Navazesh 1993), unstimulated saliva was collected, pretreated, and stored at −80 °C for analysis. The 16S rRNA gene amplicon–based microbiome analytical procedures, including microbial diversity, co-occurrence network, and metabolism prediction, are described in the Appendix. Statistical analyses of compositional and microbial load data, as well as machine learning analyses, were used to extract the signature species.

Results

Microbial Richness and Diversity Are Lower in the OSCC-OSF Group

Amplicon sequence variants (ASVs) were obtained with tag-encoded high-throughput sequencing of the 16S rRNA gene to study the diversity of bacterial populations in saliva samples. Four indices were used to evaluate alpha diversity: Pielou's evenness index, the observed ASVs, Shannon's diversity index, and Faith's phylogenetic diversity (PD; Appendix Fig. 1). Based on the Kruskal-Wallis test, the observed ASVs and Faith's PD were significantly higher in the OSF group than in the OSCC-OSF group ($P = 0.028$ and $P = 0.015$, respectively).

Stochasticity Intensified the Salivary Microbiome Assembly in OSCC-OSF

Ecologic mechanisms comprising deterministic processes (e.g., host selection, immune response) and stochastic processes (e.g., environmental factors, stress, chemical agents exposure) simultaneously influence microbiome assembly (Zaneveld et al. 2017). To evaluate their relative predominance in governing the saliva microbiome, null model–based stochasticity ratios were quantified with Bray-Curtis and weighted UniFrac distances. All mean stochasticity ratios of both distance measures in the 2 groups exceeded 50% (Fig. 1). Notably, the mean stochasticity ratios were larger in the OSCC-OSF group.

Oral Health Status and Smoking Contributed to Microbiome Composition Variations

The microbiome structures were compared between the groups with 4 distance measures. In principal coordinate analysis, the

phylogenetic distance measures, namely weighted and unweighted UniFrac distances, had more power to distinguish the OSF and OSCC-OSF groups than did the Jaccard and Bray-Curtis dissimilarities at the species level. The explained variation for the first 2 principal coordinates was higher with UniFrac distances (44.8%) than with Jaccard and Bray-Curtis distances (<26.1%; Fig. 2A). Distance-based Adonis analysis was then performed to test whether the microbiome composition variation was attributable to different participant characteristics. Because all patients in this study used betel quid, betel quid consumption was excluded from the analysis. Tests with Bray-Curtis and weighted UniFrac distance matrices revealed no association of patient age, smoking status, alcohol use, or oral health status (periodontal condition and OSF carcinogenesis) with variations in microbiome composition (Fig. 2B). However, tests with Jaccard and unweighted UniFrac distances illustrated that OSF carcinogenesis (OSF vs. OSCC-OSF) and smoking status significantly contributed to variations in the microbiome composition ($P < 0.05$).

Core Salivary Microbiomes in the 2 Groups

In total, 6,160 unique ASVs were detected in 52 samples. These ASVs, comprising approximately 99.9% of all reads, were annotated to 12 bacterial phyla, including *Firmicutes* (mean ± SD, 35.8% ± 12.8%), *Bacteroidetes* (21.9% ± 10.6%), *Proteobacteria* (21.0% ± 14.0%), *Saccharibacteria* (TM7) (7.3% ± 9.5%), *Actinobacteria* (6.0% ± 5.5%), *Fusobacteria* (5.5% ± 4.3%), and *Spirochaetes* (1.6% ± 2.6%). Regarding species annotation, 408 and 470 taxa were detected in OSF and OSCC-OSF samples, respectively. The most abundant species was unclassified *Streptococcus* sp. (OSF, 5.853% ± 3.224%; OSCC-OSF, 8.865% ± 6.563%), followed by unclassified *Neisseria* sp. (OSF, 5.547% ± 5.609%; OSCC-OSF, 6.374% ± 8.452%). *Veillonella atypica* (4.687% ± 5.810%) and *Haemophilus parainfluenzae* (5.500% ± 6.594%) ranked at the third position in the OSF and OSCC-OSF groups. Among the taxa, 51 (12.5%) and 28 (5.96%) were present with >75% prevalence in the OSF and OSCC-OSF samples. Therefore, these taxa were considered part of each group's core salivary microbiome (Appendix Fig. 2). These core species constituted most salivary microbes and accounted for average read abundances of 64.5% and 52.7% in the OSF and OSCC-OSF groups. Among the core species, 25 were common to both groups, whereas 26 and 3 were specific to the OSF and OSCC-OSF samples.

Machine Learning and Traditional Statistics Detected Signature Species in Both Groups

Multiple analysis methods were used to identify distinct microorganisms associated with oral health status in both groups. First, the linear discriminant analysis effect size (LEfSe) identified 42 species with differential abundance (linear discriminant analysis score >2) between the groups (Fig. 3A, Appendix Table 2). *Haemophilus pittmaniae, Prevotella* sp. HMT-309, and *Treponema* sp. HMT-270 were significantly more abundant in the OSCC-OSF group ($P < 0.05$). The prevalence of *H. pittmaniae* and *Prevotella* sp. HMT-309 was considerably higher in OSCC-OSF samples (Q2 prevalence) than in OSF samples (Q1 prevalence). Additionally, 39 species belonging to *Prevotella* (9 species), *Treponema* (6 species), *Selenomonas* (3 species), and 21 other genera (1 species per genus) were significantly more abundant in OSF samples ($P < 0.05$). Only 6 species—namely, *Mycoplasma faucium, Prevotella denticola, Clostridiales* sp. HMT-093, *Prevotella baroniae, Prevotella oulorum*, and *Selenomonas sputigena*—were core species (Q4 prevalence), and their prevalence in OSCC-OSF samples decreased by a larger magnitude (up to 41.0% ± 6.4%) than transient species taxa (prevalence <75%; Appendix Fig. 3).

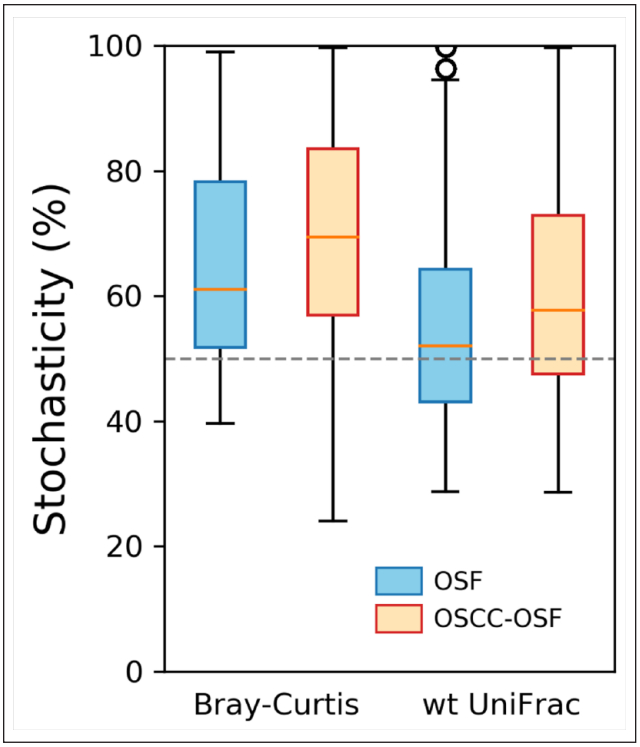


Figure 1. Null model–based stochasticity ratio of microbiome assembly based on Bray-Curtis and weighted UniFrac dissimilarities. The simulated assembly procedure was repeated 999 times. Kruskal-Wallis test: *H* statistic = 8.695, $P = 0.0032$ for Bray-Curtis dissimilarity; *H* statistic = 10.970, $P = 0.0009$ for weighted UniFrac dissimilarity. Values are presented as median (line), interquartile range (box), 1.5 interquartile range (whiskers), and outliers (circles). OSCC, oral squamous cell carcinoma; OSF, oral submucous fibrosis.

To supplement the compositional data, we converted the relative abundance into microbial loads (cells per milliliter of saliva) by multiplying the 16S rRNA copy number–corrected relative abundance by the 16S rRNA copy number–corrected quantitative polymerase chain reaction–quantified cell number (Appendix Fig. 4). The absolute microbial load data indicated that the quantities of 23 species from 15 genera were statistically different between the groups ($P < 0.05$; Fig. 3B). All 23 species corresponded to LEfSe results (Fig. 3A).

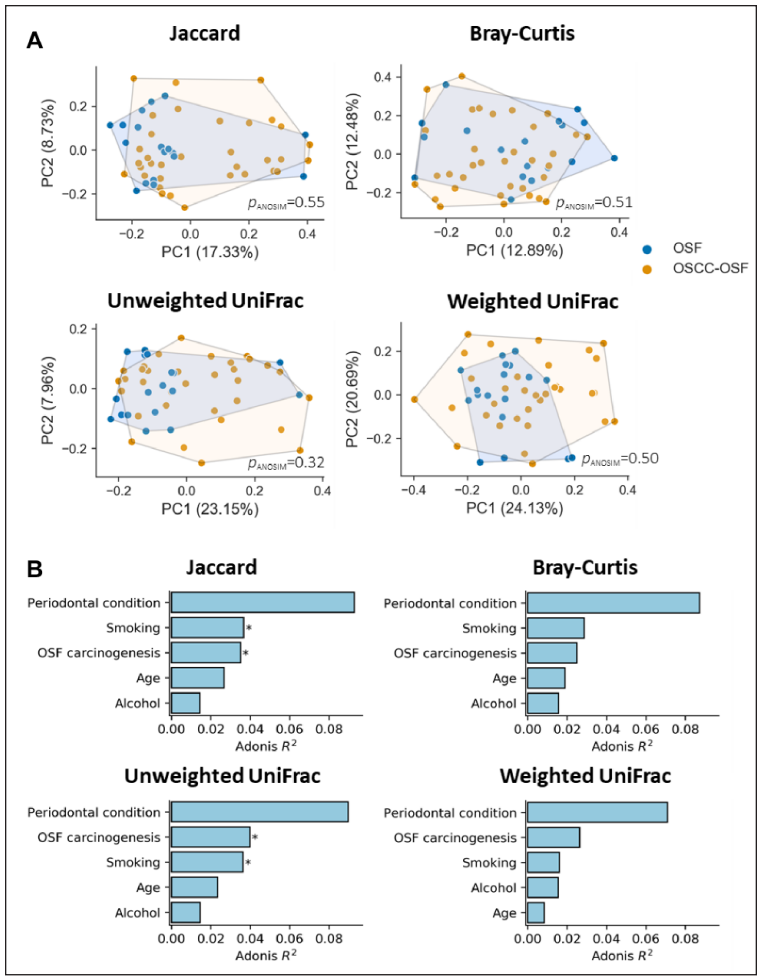


Figure 2. Beta diversity comparisons in the association with oral health status and risk factors. (**A**) Principal coordinate analysis and (**B**) Adonis analysis of Jaccard, Bray-Curtis, weighted UniFrac, and unweighted UniFrac distance metrics of salivary microbiomes in the OSF and OSCC-OSF groups. The percentage on the axes in panel A indicates the fraction of variance explained by each axis. *$P < 0.05$.

In addition, smoking status significantly affected microbiome composition (Fig. 2B). According to the Kruskal-Wallis test of absolute microbial load data, 28 species were associated with participant smoking habits, and 12 species within 8 genera were also related to OSF carcinogenesis (Appendix Fig. 5). Moreover, 11 species were related to OSF carcinogenesis. Excluding *Eubacterium saphenus*, these species were unclassified. No signature species of OSF carcinogenesis significantly differed in abundance with changing periodontal conditions. Thus, there is a distinct association between signature species and OSF carcinogenesis.

Signature species taxa were also detected with machine learning. The machine learning feature selection algorithm detected 15 species that influenced discriminability. Five of these species—namely, *Porphyromonas catoniae, Prevotella multisaccharivorax, Prevotella* sp. HMT-300, *Mitsuokella* sp. HMT-131, and *Treponema* sp. HMT-927—corresponded to those detected via statistical analysis (Fig. 3B). Meanwhile, *Dialister micraerophilus* and *Mollicutes* sp. HMT-504 were also detected through LEfSe (Fig. 3A). The remaining 8 species, which were detected with the machine learning feature selection algorithm but not statistical methods, were *Clostridiales* sp. HMT-876, *Corynebacterium durum, Gracilibacteria* sp. HMT-871, *Megasphaera* sp. HMT-123, *Mobiluncus mulieris, Negativicutes* (unclassified), *Peptostreptococcaceae* (unclassified), and *Selenomonas* sp. HMT-478.

Artificial Intelligence Achieved High Discriminability to Detect OSF-Related Malignancy

The effectiveness of 23 signature species in discriminating the OSF and OSCC-OSF groups was evaluated by calculating the area under the receiver operating characteristic curve (AUC) according to the relative abundance and microbial load. In addition to cross-validation based on an exploratory data set, the results were evaluated with an independent salivary microbiome data set (Table). The average AUCs were 0.68 and 0.56 for the exploratory and validation data sets, respectively. *Treponema* sp. HMT-927, which was attributable to oral health and smoking factors (Appendix Fig. 4), exhibited the highest discriminability in the exploratory data set ($AUC_{relative}$ = 0.748 and $AUC_{absolute}$ = 0.758) and high performance in the validation data set ($AUC_{relative}$ = 0.709 and $AUC_{absolute}$ = 0.682). The best-performing species was *Absconditabacteria* (SR1) sp. HMT-874 ($AUC_{relative}$ = 0.755 and $AUC_{absolute}$ = 0.809 for the validation data set), which is a taxon in the candidate phyla radiation group. *Absconditabacteria* (SR1) sp. HMT-874 was a signature oral health factor. No statistical significance was observed between AUCs derived from the relative abundance and absolute microbial load for either exploratory (H = 0.008, P = 0.923) or validation (H = 0.088, P = 0.767) data sets.

With the combination of species features and host clinical and lifestyle features, the machine learning analysis achieved high discriminability. The random-forest algorithm trained with an input of species taxa detected with the feature selection, Faith's PD, and lifestyle (current cigarette smoking) most efficiently discriminated the OSF and OSCC-OSF groups with 85.1% mean 5-fold cross-validation accuracy and an AUC of 0.88 (Appendix Fig. 6). The results in Figure 2B and Appendix Figure 1 support the importance of the detected features, Faith's PD, and smoking status.

Microbial Networks Collapsed in the OSCC-OSF Group

In species-species co-occurrence network analysis, patients with OSF had approximately 3-fold more complex microbial

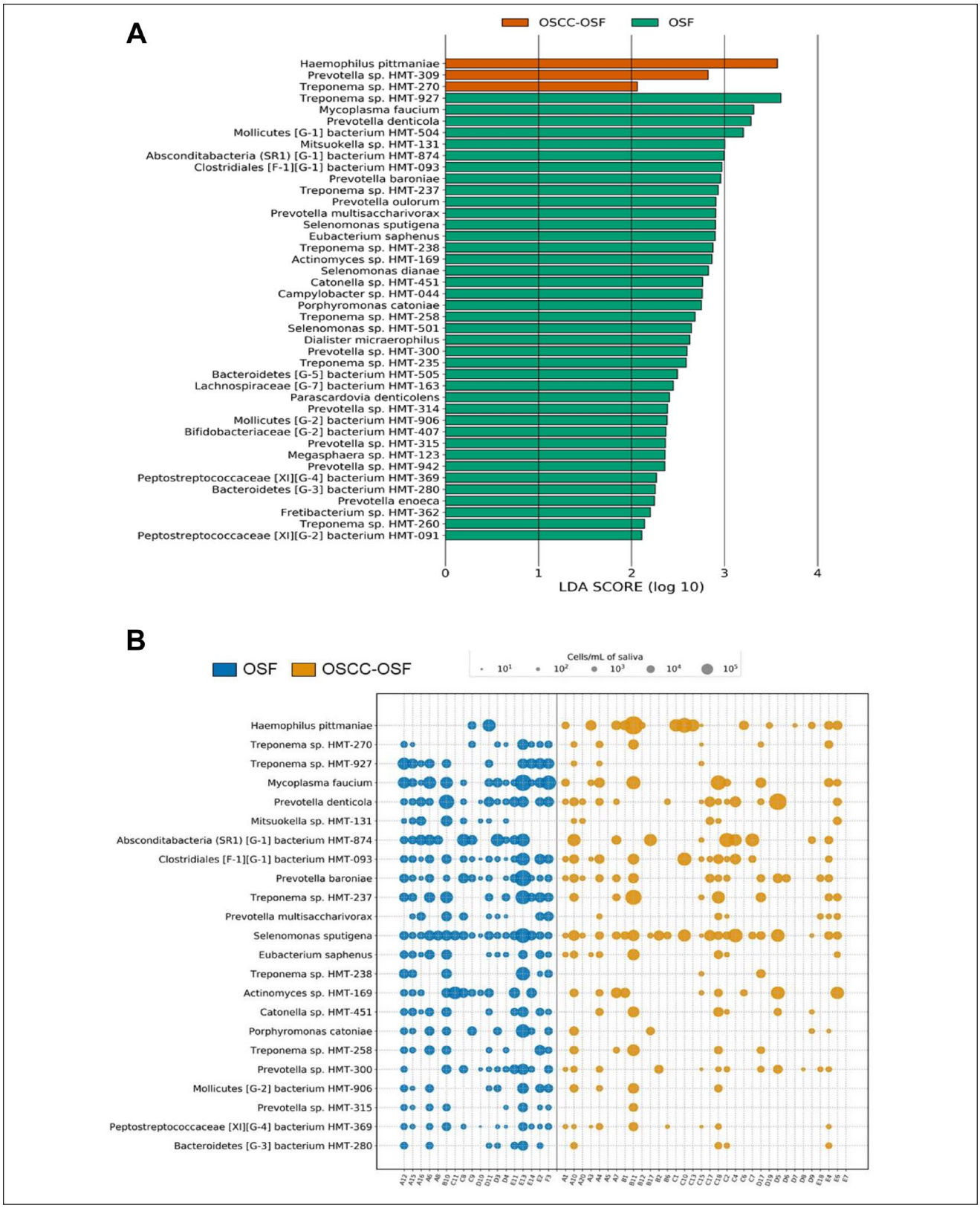


Figure 3. Distinct species between the OSF and OSCC-OSF groups detected via relative abundance and absolute microbial load. (**A**) LDA effect size ($\alpha = 0.05$, $\log_{10}$ LDA score >2.0) revealed 39 species (green) with greater relative abundance in the OSF group and 3 species (red) with greater relative abundance in the OSCC-OSF group. The relative abundance and prevalence of these species are listed in Appendix Table 2. (**B**) The Kruskal-Wallis test ($\alpha = 0.05$, effect size $E_R^2 > 0.06$, prevalence of a species >33% in at least 1 group) for absolute microbial loads detected 23 species with significantly different quantities between the groups. LDA, linear discriminant analysis; OSCC, oral squamous cell carcinoma; OSF, oral submucous fibrosis.

Table. Area Under the Receiver Operating Characteristic Curve of Candidate Signature Species Based on the Relative Abundance and Absolute Microbial Load.

			Exploratory Data Set		Validation Data Set	
No.	Species	Core Species	Absolute	Relative	Absolute	Relative
1	*Treponema* sp. HMT-927		0.758	0.748	0.682	0.709
2	*Mycoplasma faucium*	OSF	0.753	0.737	0.645	0.645
3	*Prevotella denticola*	OSF	0.725	0.730	0.505	0.550
4	Peptostreptococcaceae sp. HMT-369		0.724	0.684	0.595	0.577
5	*Porphyromonas catoniae*		0.703	0.696	0.686	0.632
6	*Prevotella* sp. HMT-315		0.703	0.704	0.727	0.609
7	Clostridiales sp. HMT-093	OSF	0.697	0.700	0.500	0.536
8	*Eubacterium saphenus*		0.696	0.690	0.555	0.536
9	*Catonella* sp. HMT-451		0.693	0.694	0.527	0.491
10	*Treponema* sp. HMT-237		0.680	0.658	0.555	0.600
11	*Selenomonas sputigena*	OSF	0.676	0.724	0.609	0.564
12	*Prevotella multisaccharivorax*		0.675	0.679	0.464	0.436
13	*Haemophilus pittmaniae*		0.673	0.672	0.573	0.591
14	*Prevotella baroniae*	OSF	0.673	0.712	0.382	0.364
15	*Prevotella* sp. HMT-300		0.672	0.685	0.582	0.618
16	*Actinomyces* sp. HMT-169		0.669	0.682	0.364	0.364
17	Absconditabacteria (SR1) sp. HMT-874		0.667	0.658	0.809	0.755
18	*Treponema* sp. HMT-270		0.660	0.644	0.509	0.491
19	Mollicutes sp. HMT-906		0.660	0.658	0.418	0.409
20	*Mitsuokella* sp. HMT-131		0.644	0.660	0.555	0.509
21	Bacteroidetes sp. HMT-280		0.644	0.630	0.550	0.505
22	*Treponema* sp. HMT-238		0.642	0.632	0.527	0.518
23	*Treponema* sp. HMT-258		0.639	0.644	0.609	0.618

The exploratory and validation data sets were both used to evaluate diagnostic performance.
OSF, oral submucous fibrosis.

interactions (7,173 edges connecting 362 nodes) than those with OSCC-OSF (2,695 edges connecting 351 nodes). Figure 4 illustrates completely different network structures of signature species between the salivary ecosystem groups. The OSF group possessed high connectivity with more positive relations (solid lines) than negative relations (dotted lines), indicating a distinct species co-occurrence pattern (Fig. 4A). In addition to *E. saphenus, M. faucium, Catonella* sp. HMT-451, and *Clostridiales* sp. HMT-093, several signature species, including *Treponema* (HMT-927, HMT-258, HMT-237, and HMT-238) and *Prevotella* (*P. denticola* and HMT-315), served as central hubs that were highly connected with positive edges (solid lines) in the network. Two hub species, *Neisseria* sp. (unclassified) and *Oribacterium* sp. (unclassified), were correlated with other species with mostly negative edges (dotted lines). By contrast, the corresponding species co-occurrence pattern was not present in patients with OSCC-OSF (Fig. 4B).

Metabolism Shifted in Salivary Microbiomes

To compare biochemical functions of saliva microbiomes in the 2 groups, LEfSe was used to identify quantitative differences in predicted metabolic pathways. Although numerous signature species were detected in the 2 groups, only 9 metabolic pathways were quantitatively discriminable (Appendix Fig. 7). Six pathways were more closely associated with OSF samples, and they included pathways for L-ornithine, L-histidine, pyrimidine deoxyribonucleotide, and nicotinamide adenine dinucleotide synthesis as well as formaldehyde oxidation and assimilation. In the OSCC-OSF group, 3 metabolic pathways were identified: 1,4-dihydroxy-6-naphthoate (an intermediate of vitamin K_2 synthesis), *S*-adenosyl-L-methionine (SAM), and norspermidine synthesis.

Discussion

Consistent with previous studies (Hernandez et al. 2017; Lee et al. 2017), this study revealed that habitual smoking considerably alters bacterial diversity in the saliva of male patients with OSF who chew betel quid. Although the host's health status is crucial to salivary bacterial diversity (Wade 2013; Meuric et al. 2017), there is no consensus concerning the changes of bacterial diversity during cancer formation. One study identified significantly higher bacterial diversity in the cancer group than in the healthy group (Zhao et al. 2017), whereas a recent report revealed that modifications in the oral microbiome may mirror OSF carcinogenesis (Zhong et al. 2020), supporting our results. Another study uncovered no significant difference in overall bacterial richness and diversity between the control and precancerous lesion groups (Lee et al. 2017). Together, these results suggested the various effects of health status on the changes in salivary bacterial diversity.

Similar to observations in diabetes mellitus and Crohn's disease (Dey et al. 2013), stochastic changes in the salivary microbiome are profound, and environmental factors are considered the main determinants of the oral microbiome structure

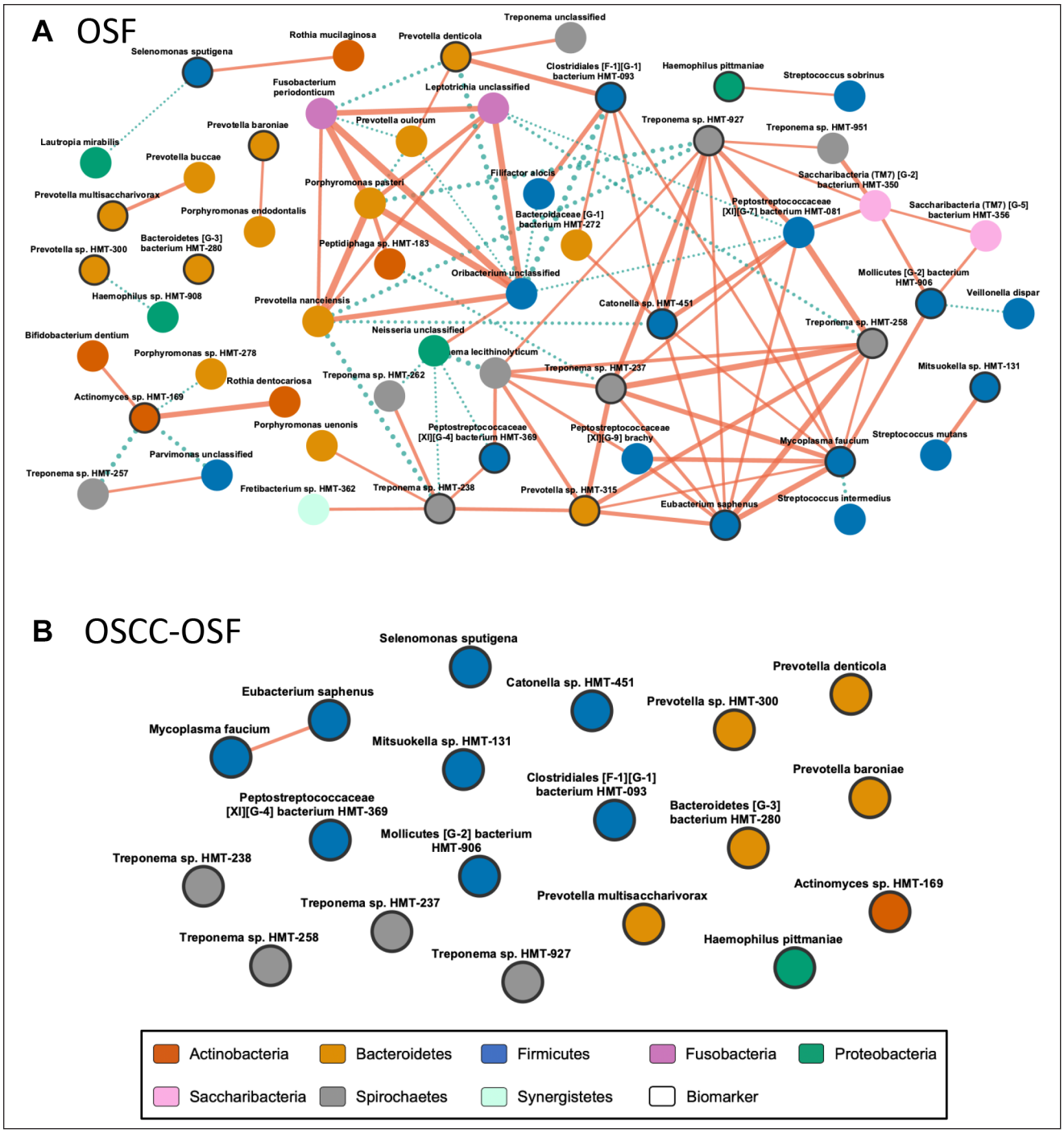


Figure 4. Network analysis of salivary microbiomes in (**A**) OSF and (**B**) OSCC-OSF samples via SparCC. Species identified by relative abundance and absolute microbial load and their first neighbors with $r_{SparCC} > 0.6$ (solid lines) and $r_{SparCC} < -0.6$ (dotted lines) are displayed. The node color denotes phylum. OSCC, oral squamous cell carcinoma; OSF, oral submucous fibrosis.

(Stahringer et al. 2012). The community instability associated with cancer formation in the OSCC-OSF group was likely caused by stochastic perturbations attributable to exposure to carcinogens, smoking habits, and microenvironmental changes (e.g., oral submucous fibrosis; Zaneveld et al. 2017). Because disease progression often reduces the host's ability to regulate microbiota composition, the role of stochasticity in determining the salivary microbiome structure was increased in the OSCC-OSF group. Bacteria coexist in complex interaction webs, and interactions within these webs affect the species involved. Meanwhile, perturbations may be associated with disease (Faust and Raes 2012). Other evidence (Appendix Fig. 2, Fig. 4) was consistent with the stochasticity-dominating salivary microbiome in the dysbiotic state.

Several oral species have been associated with OSCC (Mager et al. 2005), and a link between biofilm-induced periodontal diseases and oral cancer has been proposed (Meurman and Bascones-Martinez 2011), in line with our findings. The 2 culturable species, namely *P. catoniae* and *P. multisaccharivorax*, have been identified as oral pathogens (de Lillo et al. 2004; Pati et al. 2011). The other signature species were uncultured, and limited information concerning their role in oral carcinogenesis is available. Despite the lack of a significant difference between species abundance and the periodontal conditions (Appendix Fig. 5), *Prevotella, Mitsuokella*, and *Treponema* species are commonly detected in the subgingival plaques of patients with poor oral hygiene and chronic periodontitis, and they are possibly associated with oral cancer (Chattopadhyay et al. 2019; Zhong et al. 2020).

The altered microbial compositions, including the collapse of the species co-occurrence network, may contribute to a metabolic shift, especially dysregulating polyamine and SAM synthesis. The nonproteinogenic amino acid ornithine can be metabolized anaerobically by numerous bacterial species (Fonknechten et al. 2009). Ornithine in saliva can originate from foods or supplements, the anaerobic degradation of sugars (Xu et al. 2015) and arginine (Noens et al. 2015) by bacteria, and host cells (Nowotarski et al. 2013). In mammalian cells, ornithine can facilitate the synthesis of putrescine and other polyamines through ornithine decarboxylase activity, the pathway of which is connected to the SAM-mediated synthesis of spermidine and spermine (Roje 2006). These study results suggest that oral carcinogenesis in patients with OSF can involve dysregulated ornithine, arginine, and polyamine synthesis in the salivary microbiome. Increased polyamine levels have been associated with numerous cancers, such as breast, colon, lung, prostate, and skin (Nowotarski et al. 2013). SAM is the principal biological methyl donor in transmethylation reactions that are fundamental to sustaining epigenetic modifications, and it is strongly correlated with carcinogenesis (Lu and Mato 2012; Irimie et al. 2018). Thus, the high potential of SAM synthesis in the OSCC-OSF salivary microbiome might be conducive to the host's oral carcinogenesis. Norspermidine is generated only by bacteria and plants (Gerner and Meyskens 2004), and menaquinone (vitamin K_2) has been observed to have anticancer activities in animal and cell studies (Vinjamuri et al. 2020). More studies regarding the relationships of these metabolites to oral carcinogenesis are warranted.

Finally, our functional prediction analysis suggested chronic inflammation in patients with OSF. L-Histidine is not synthesized in humans, but it can be converted into histamine by histidine decarboxylase, resulting in inflammation (Branco et al. 2018). Formaldehyde, a potent carcinogen (Hopkinson and Schofield 2018), can be produced endogenously by the host, bacteria, or host-bacteria interactions (Chen et al. 2016). During the inflammatory process in OSF formation, macrophages and neutrophils produce reactive oxygen species and formaldehyde (Burgos-Barragan et al. 2017). Inadequate formaldehyde detoxification by formaldehyde oxidation and assimilation in the salivary microbiomes of patients with OSCC-OSF was likely related to oral cancer formation. Because most salivary species were transient, bacterial communities may be compositionally different but functionally similar, suggesting that function (metagenomic)–based analysis is more efficient than identity-based analysis for identifying the relationship between the salivary microbiome and oral carcinogenesis. However, because microbiome functions were in silico predicted in this study, metabolomics analyses of saliva samples are required for future validation.

This study was limited by its relatively small sample size, and all samples were from Taiwanese participants. Because the human microbiome is significantly influenced by diet and personal habit, further research in different countries is needed. Although we identified several species related to carcinogenesis in the OSF background, shotgun metagenomics or culturomics studies are required to understand the detailed mechanism of salivary microbiome and oral cancer formation.

Despite these considerations, our study, for the first time, provided microbial ecological perspectives and mechanistic insights into the alterations of the salivary microbiome during the malignant transformation of OSF in male betel quid chewers. These novel findings, with the OSF-OSCC–associated species as biomarkers detected in this study, can also provide references for precision medicine to predict cancer formation in high-risk populations.

Author Contributions

M.Y. Chen, contributed to conception, design, data acquisition, and interpretation, drafted the manuscript; J.W. Chen, contributed to conception, data analysis, and interpretation, drafted the manuscript; L.W. Wu, contributed to conception, design, and data interpretation, critically revised the manuscript; K.C. Huang, J.Y. Chen, contributed to data acquisition, drafted the manuscript; W.S. Wu, C.J. Shih, K.N. Tsai, W.T. Hsieh, Y.H. Ho, contributed to data analysis, critically revised the manuscript; W.F. Chiang, contributed to design and data acquisition, critically revised the manuscript; T.Y. Wong, contributed to conception, critically revised the manuscript; J.H. Wu, Y.L. Chen, contributed to conception and design, critically revised the manuscript. All authors gave final approval and agree to be accountable for all aspects of the work.

Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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Data Availability

Raw sequence data have been deposited into the European Nucleotide Archive at EMBL-EBI under accession PRJEB37501. The identified species in all samples and their relative abundances are detailed in the Appendix.

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